Memory

A-MEM Mem0 Memory OS NEMORI

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1. A-MEM: Agentic Memory for LLM Agents
2. Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory
3. Memory OS of AI Agent
4. NEMORI: What Deserves Memory
5.
6. Agent

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1. A-MEM: Agentic Memory for LLM Agents

	A-MEM: Agentic Memory for LLM Agents Wujiang Xu (Rutgers), Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, Yongfeng Zhang **NeurIPS 2025** (The Thirty-ninth Annual Conference on Neural Information Processing Systems) **CCF-A** ( )
arXiv	2502.12110 https://github.com/WujiangXu/A-Mem

A-MEM **Zettelkasten** LLM Agent

1. Note Construction - $m_i = \{c_i, t_i, K_i, G_i, X_i, e_i, L_i\}$ - LLM - LLM

2. Link Generation - top-k - LLM - "box" Zettelkasten

3. Memory Evolution - - -

Figure 1: A-MEM

4. Memory Retrieval - - box

A-MEM *Note Construction Link Generation Memory Evolution Memory Retrieval*

LoCoMo 6

		Multi-Hop (F1)	Temporal (F1)	Single-Hop (F1)	Open Domain (F1)
GPT-4o-mini	A-MEM	**27.02**	**45.85**	**44.65**	12.14
GPT-4o-mini	MemGPT	26.65	25.52	41.04	9.15
GPT-4o	A-MEM	**32.86**	**39.41**	48.43	17.10
Qwen2.5-1.5B	A-MEM	**18.23**	**24.32**	**23.63**	**16.48**
Llama 3.2-3B	A-MEM	**17.44**	**26.38**	**28.14**	12.53

- **Multi-Hop 2**
- ranking 1

DialSim - F1=3.45 (vs LoCoMo 2.55, MemGPT 1.18 35%/192%)

GPT-4o-mini

	Multi-Hop F1	Temporal F1	Open Domain F1
w/o LG & ME (+)	9.65	24.55	7.77
w/o ME ( )	21.35	31.24	10.13

	Multi-Hop F1	Temporal F1	Open Domain F1
Full A-MEM	**27.02**	**45.85**	**12.14**

- 1,200 tokens MemGPT/LoCoMo 16,900 tokens **85-93%** - < $0.0003 API - 100 3.70µs

- 1K→1M 0.31µs 3.70µs - O(N)

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2. Mem0: Building Production-Ready AI Agents

	Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, Deshraj Yadav **arXiv** (2025-04-28) Mem0
arXiv /	2504.19413 https://mem0.ai

Mem0

2.1 Mem0 **Extraction Phase** - (m_{t-1}, m_t) + S + m - LLM $\Omega = \{\omega_1, ..., \omega_n\}$

Update Phase - top-s - LLM Tool Call - **ADD** - **UPDATE** - **DELETE** - **NOOP**

2.2 Mem0^g

- $G = (V, E, L)$
- =
- = (v_s, r, v_d)
- →
-
- +

Mem0 *Extraction* *Update*

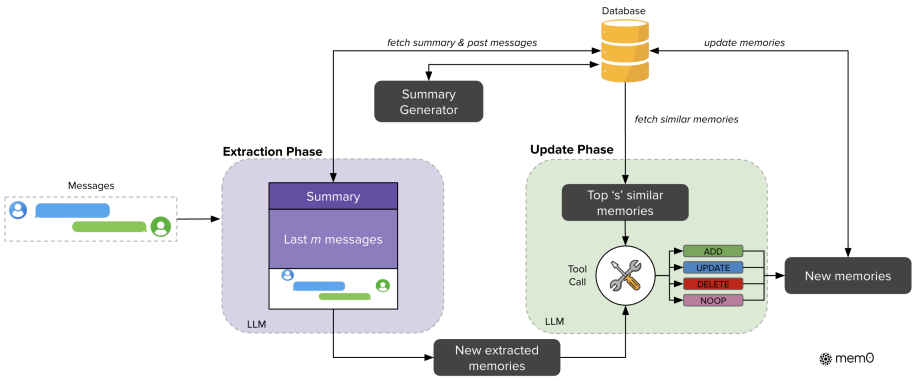

Figure 2: Mem0

LoCoMo GPT-4o-mini

	Single-Hop (J)	Multi-Hop (J)	Temporal (J)	Open Domain (J)	Overall J
A-Mem*	39.79	18.85	49.91	54.05	48.38
LangMem	62.23	47.92	23.43	71.12	58.10
Zep	61.70	41.35	49.31	**76.60**	65.99
OpenAI	63.79	42.92	21.71	62.29	52.90
Mem0	**67.13**	**51.15**	55.51	72.93	66.88
Mem0^g	65.71	47.19	**58.13**	75.71	**68.44**

	Search p95 (s)	Total p95 (s)	Overall J
Full-context	-	17.117	**72.90**
Mem0	**0.200**	**1.440**	66.88
Mem0^g	0.657	2.590	68.44
A-Mem	1.485	4.374	48.38
LangMem	59.82	60.40	58.10

- Mem0 Full-context **p95** **91% token** **90%+**
- J Full-context 5

Token - Mem0 7K tokens vs Full-context 26K, Zep >600K - Mem0^g 14K tokens

- Mem0 Single-Hop Multi-Hop - Mem0^g Temporal Open Domain -

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3. Memory OS of AI Agent

arXiv	Memory OS of AI Agent Jiazheng Kang ( ), Mingming Ji (AI Lab), Zhe Zhao (AI Lab), Ting Bai ( ) **EMNLP 2025** (Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing) **CCF-B** ( ) 2506.06326 https://github.com/BAI-LAB/MemoryOS

MemoryOS

Memory Storage

STM	(dialogue pages)	FIFO
MTM	-	heat
LPM	/Agent	User Persona + Agent Persona

Memory Updating - **STM→MTM** FIFO - **MTM** + F_{score} - **Heat** $Heat = \alpha \cdot N_{visit} + \beta \cdot L_{interaction} + \gamma \cdot R_{recency}$ - heat heat LPM - **MTM→LPM** heat τ - **LPM** 90 Agent

Memory Retrieval - STM - MTM top-m → top-k - LPM top-10 /Agent

Response Generation - STM + MTM + LPM prompt

MemoryOS STM/MTM/LPM

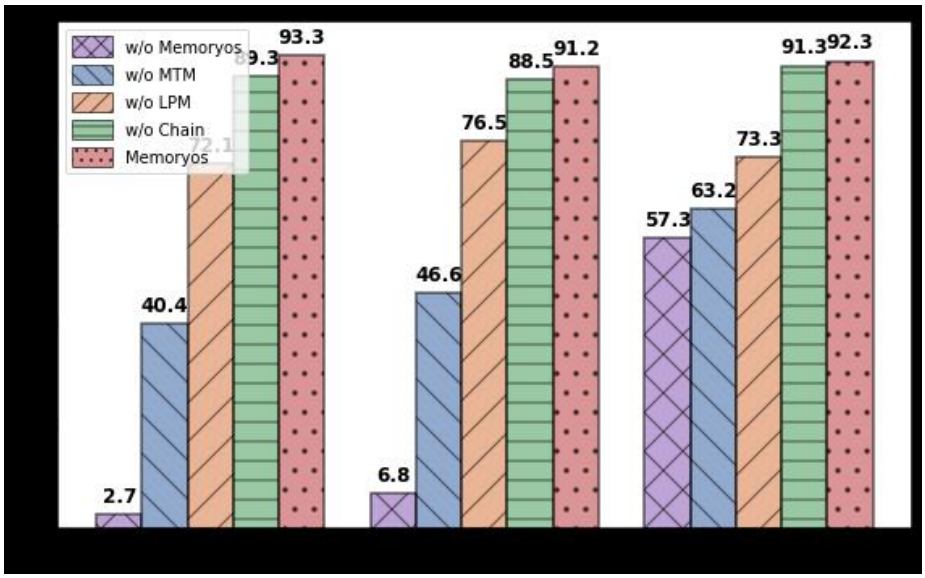

Figure 3: MemoryOS

GVD GPT-4o-mini

	Accuracy ↑	Correctness ↑	Coherence ↑
TiM	84.5	78.8	90.8
MemoryBank	78.4	73.3	91.2
MemGPT	87.9	83.2	89.6
A-Mem	90.4	86.5	91.4
MemoryOS	**93.3**	**91.2**	**92.3**
Improvement	+3.2%	+5.4%	+1.0%

LoCoMo GPT-4o-mini

	Single Hop F1	Multi Hop F1	Temporal F1	Open Domain F1	Avg Rank (F1)
TiM	16.25	18.43	8.35	23.74	3.8
MemoryBank	5.00	9.68	5.56	6.61	5.0
MemGPT	26.65	25.52	9.15	41.04	2.2
A-Mem*	22.61	33.23	8.04	34.13	3.0
MemoryOS	**35.27**	**41.15**	**20.02**	**48.62**	**1.0**
Improvement	+32.35%	+23.83%	+118.80%	+18.47%	-

- **F1 49.11% BLEU-1 46.18%**

	Avg Tokens	Avg LLM Calls	Avg F1
MemoryBank	432	3.0	6.84
TiM	1,274	2.6	18.01
MemGPT	16,977	4.3	29.13
A-Mem*	2,712	13.0	26.55
MemoryOS	3,874	**4.9**	**36.23**

- MTM → - LPM → - Chain → - MemoryOS →

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4. NEMORI: What Deserves Memory

arXiv	What Deserves Memory: Adaptive Memory Distillation for LLM Agents Wenquan Ma (/), Jiayan Nan ( ), Wenlong Wu ( ), Yize Chen ( ) **arXiv** (v4: 2026-04-16) "anonymous reviewers" 2026 2508.03341 https://github.com/nemori-ai/nemori

NEMORI **Predictive Coding Theory** " "

Priors

Structure Prior		episode
Representation Prior	allocentric	egocentric
Distillation Prior	=	

4.1 Episodic Memory Integration

- **Local Message Partitioning** LLM episode
- **Narrative Episode Generation** episode + cue
- **Associative Memory Integration** episode

4.2 Semantic Knowledge Distillation

- **Anticipatory Schema Synthesis** episode
- **Prediction Error Distillation** **insight**
- **Agnostic Knowledge Consolidation** new / merge / conflict

- `Evoke()` `Consolidate()` - DB + / - A-MEM MemoryOS

NEMORI

Figure 4: NEMORI

NEMORI PDF pdfimages Episodic Memory Integration Semantic Knowledge Distillation

LoCoMo

GPT-4.1-mini

	Temporal (LLM)	Multi-Hop (LLM)	Single-Hop (LLM)	Open Domain (LLM)	**Avg LLM**
Full Context	74.2	77.2	86.9	56.6	80.6
LangMem	50.8	71.0	84.5	59.0	73.4
Mem0	56.9	68.2	71.4	47.9	66.3
A-MEM	66.7	55.7	64.0	37.5	61.4
MemoryOS	37.7	62.4	68.9	60.4	60.6
NEMORI	**77.3**	74.8	**87.0**	56.3	**80.8**
vs strongest memory	*+15.9%*	*+5.4%*	*+3.0%*	*-6.8%*	*+10.1%*

GPT-4o-mini

	Avg LLM
Full Context	72.3
Mem0	61.3
MemoryOS	54.5

	Avg LLM
A-MEM	52.5
NEMORI	**73.0**
vs strongest memory	*+19.1%*

- NEMORI **Full Context** 80.8 vs 80.6 / 73.0 vs 72.3
- **Temporal Reasoning** GPT-4.1-mini +15.9% over best memory baseline

LoCoMo, GPT-4o-mini

	LLM Calls	Tokens	Avg LLM
Mem0	1,602	1,693k	61.3
A-MEM	1,176	1,149k	52.5
MemoryOS	1,016	527k	54.5
NEMORI	**373**	**323k**	**73.0**
Improvement	*-59.5%*	*-38.7%*	

- token 2,745 vs Full Context 23,653 **88%**
- 3,053ms vs Full Context 5,806ms **47%**

A-MEM MemoryOS

		LLM Score	Memory Tokens
A-MEM	Raw Messages (P)	52.5	397K
A-MEM	**NEMORI Semantic (K)**	50.9	142K (**-64.3%**)
MemoryOS	Raw Messages (P)	54.6	405K
MemoryOS	**NEMORI Semantic (K)**	54.0	190K (**-53.1%**)

- 45-64% ±4% +1.9% ~ +6.1%

LongMemEvalS 105K tokens 10x LoCoMo

	Full Context (101K tok)	NEMORI (3.7-4.8K tok)
GPT-4o-mini	55.0	**64.2** (+16.7%)
GPT-4.1-mini	65.6	**74.6** (+13.7%)

- NEMORI

- episodic -11.0% (gpt-4o-mini) - semantic -25.1% (gpt-4o-mini) - vs +25.0% - 20 -4~6%

5.

5.1

	A-MEM	Mem0	Memory OS	NEMORI
	Zettelkasten	+	OS	
		+		Episode→Semantic
	+	ADD/UPDATE/DELETE/NOOP	(STM/MTM/LPM) Heat-based LFU/FIFO	+
	+ Box	+	Segment→Page	Episodic + Semantic
/			90 + KB	
				+

5.2 LoCoMo

		Multi-Hop	Temporal	Single-Hop	Open Domain	
A-MEM	**NeurIPS'25**	F1=27.0	F1=45.9	F1=44.7	F1=12.1	Multi-Hop 2x
Mem0	arXiv	J=51.2	J=55.5	J=67.1	J=72.9	91%
Memory OS	**EMNLP'25**	F1=41.2	F1=20.0	F1=35.3	F1=48.6	+49% F1
NEMORI	arXiv	J=74.8	**J=77.3**	**J=87.0**	J=56.3	Full Context

F1/BLEU vs LLM-as-Judge

5.3

	LLM Calls	Token /	p95	1M
A-MEM	~13	~2,500	4.4s	**3.7µs**
Mem0		7K (base) / 14K (graph)	**1.4s**	-
Memory OS	4.9	3,874	-	-
NEMORI	**373** ( )	**323K** ( )	3.1s	-

5.4

	/	CCF	
A-MEM	NeurIPS 2025	**CCF-A**	
Mem0	arXiv		(Mem0.ai)
Memory OS	EMNLP 2025	**CCF-B**	
NEMORI	arXiv		

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6. Agent

Agent Agent

6.1

```
Agent    =    +    +
                ↑              ↑              ↑
              Memory  ↔   Distillation  ↔ Reasoning
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- **A-MEM Memory OS**
- **Mem0**
- **NEMORI** ——“ ”

6.2

A-MEM	——
Mem0	——ADD/UPDATE/DELETE
Memory OS	**+ Heat** —— STM→LTM
NEMORI	—— ” → ”

6.3

Agent Memory

1. A-MEM
2. Memory OS OS / /
3. Mem0 token +
4. NEMORI ” ”

NEMORI —— A-MEM Memory OS ” + ”

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